

Lightning Talk: Three Project Proposals

1

Vehicle Accident Severity

Classification · 4 classes

2

Food & Nutrition Category

Classification · multi-class

3

Customer Churn Prediction

Classification · binary

PROPOSAL 1: PREDICTING VEHICLE ACCIDENT SEVERITY

CAPSTONE PHASE 1: TOPIC EVALUATION

⚠️ PROBLEM & GOAL

Predict early estimate of accident impact (Severity 1–4) for responders and planners. Goal: identify high-severity factors using multiclass classification (Non-NLP).

📄 DATASET METADATA

Total Rows: 7,728,394

Features: 46 columns (13 numeric / 20 categorical)

Task Type: Ordinal Multiclass

Duplicate Rows: Zero Detected

⚙️ PROPOSED APPROACH

- Exclude free-text Description (non-NLP)
- Engineer time features from timestamps
- Compare RF / XGBoost vs Logistic baseline
- Evaluate via Macro F1 due to imbalance

MISSINGNESS RISK

End_Lat/Lng: 44.0%

Precipitation(in): 28.5%

OUTLIER IQR FLAGS

Visibility(mi): 1.48M

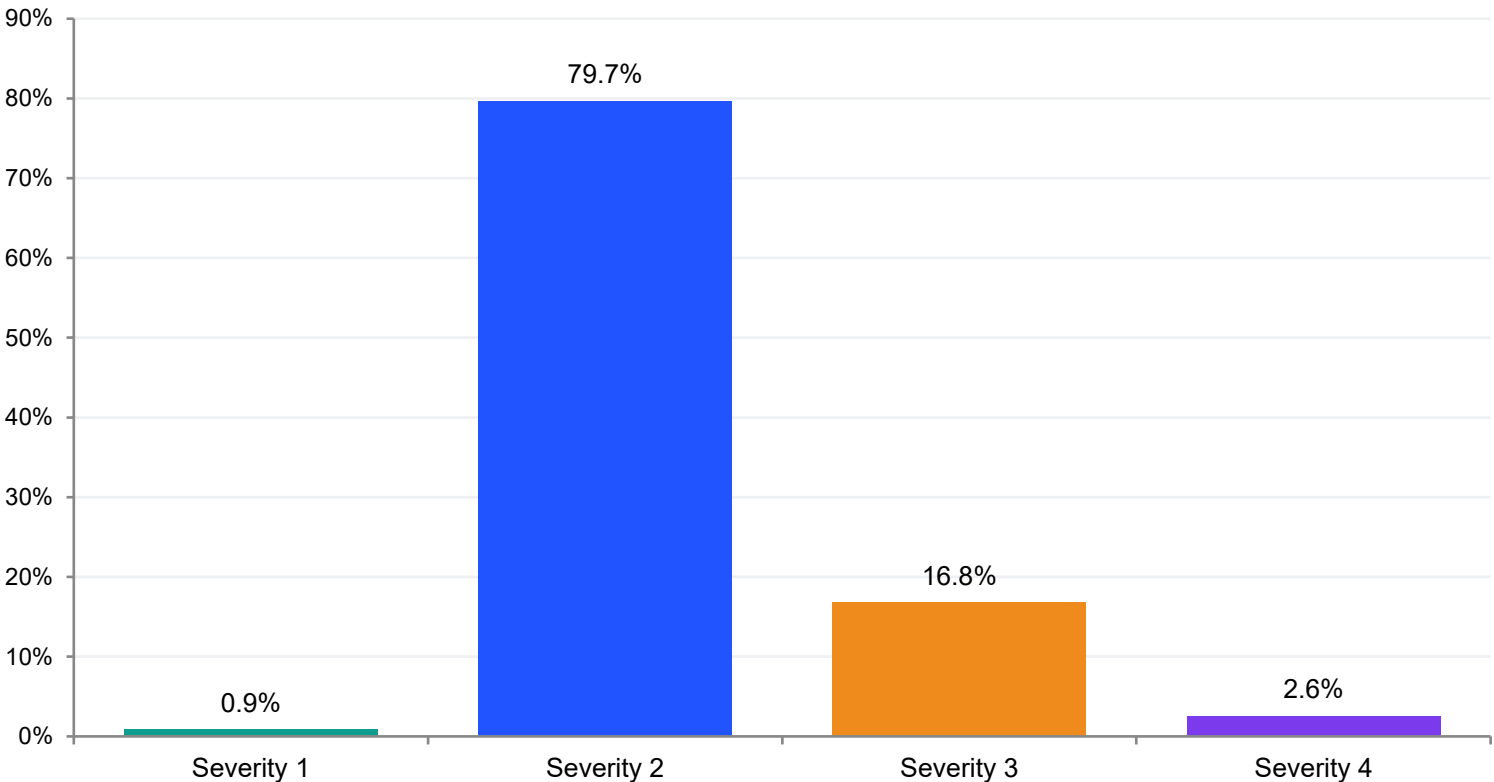
Distance(mi): 964K

LINEAR SIGNAL

Weak (max $|r|$ = 0.11)

Requires non-linearity

Target Distribution: Severity (1–4)



Sourced from executed notebook EDA (Kaggle: US Accidents)

PROPOSAL 2: FOOD / NUTRITION CLASSIFICATION

CAPSTONE PHASE 1: TOPIC EVALUATION

⚠️ PROBLEM & GOAL

Nutrition/health apps need to auto-classify food items from incomplete label data. Goal: predict food_category from nutrition facts (calories, macros, micros) — multiclass classification (Non-NLP).

📄 DATASET METADATA

Total Rows: 40,000

Features: 24 columns (15 numeric / 9 categorical)

Task Type: Multiclass (241 classes — high cardinality)

Duplicate Rows: Zero Detected

⚙️ PROPOSED APPROACH

- Exclude free-text ingredients / food_name (non-NLP)
- Engineer macro-ratio features (fat/carb/sugar %)
- Group rare categories into an 'other' bucket
- Compare RF/XGBoost vs Logistic baseline

MISSINGNESS RISK

vitamin_c_mg: 42.7%

brand_name: 24.4%

OUTLIER IQR FLAGS

vitamin_c_mg: 3,953

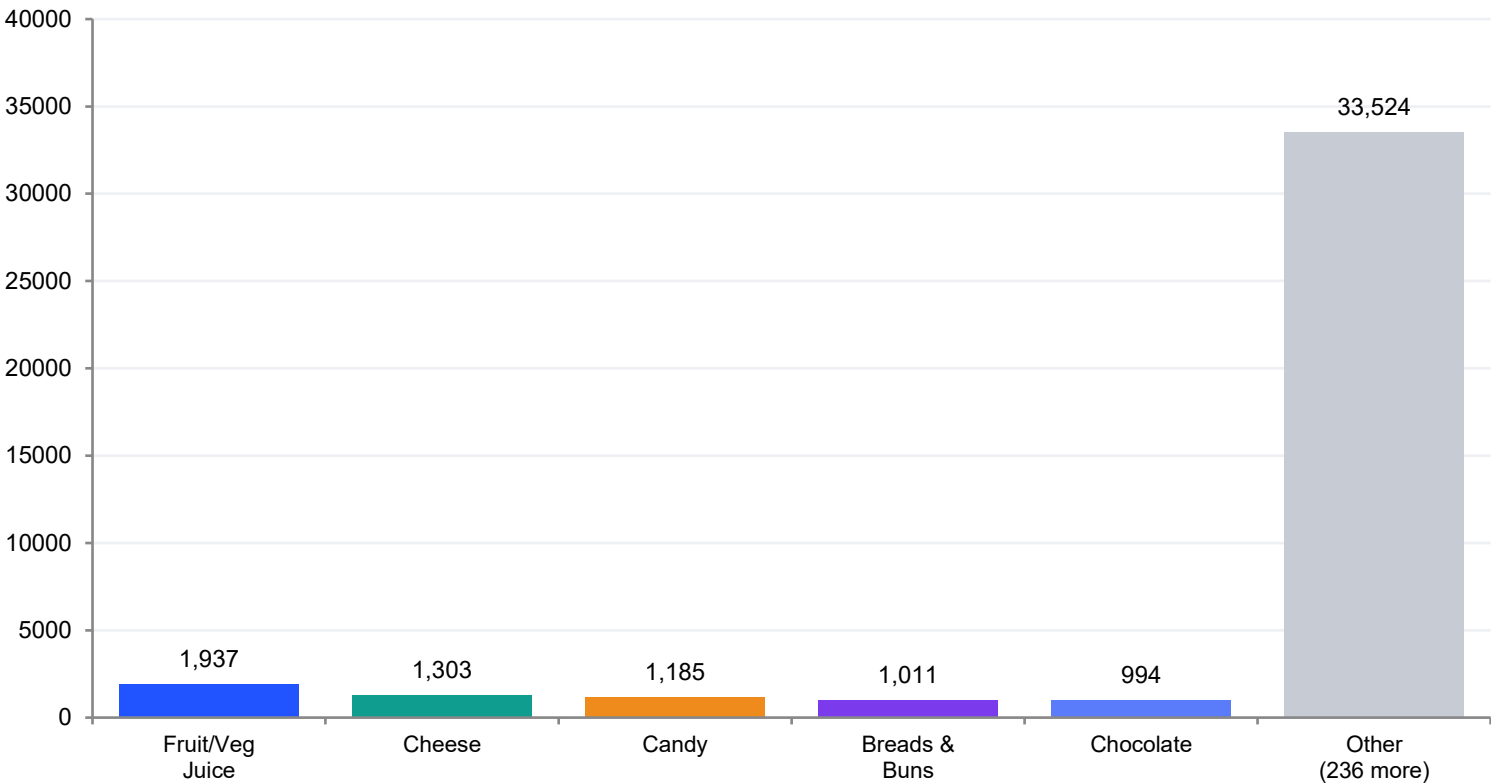
sugar_g: 3,575

STRONGEST FEATURE CORR.

sugar_g ↔ carbs_g: r=0.82

calories ↔ fat_g: r=0.60

Target Distribution: food_category (top classes)



241 total categories, 39,954/40,000 labeled — severely long-tailed (top 5 = 16% of labeled rows)

PROPOSAL 3: CUSTOMER CHURN PREDICTION

CAPSTONE PHASE 1: TOPIC EVALUATION

⚠️ PROBLEM & GOAL

Subscription businesses lose recurring revenue when customers churn, often too late to intervene. Goal: predict binary churn from account/billing/usage features so retention teams can act early (Non-NLP).

📄 DATASET METADATA

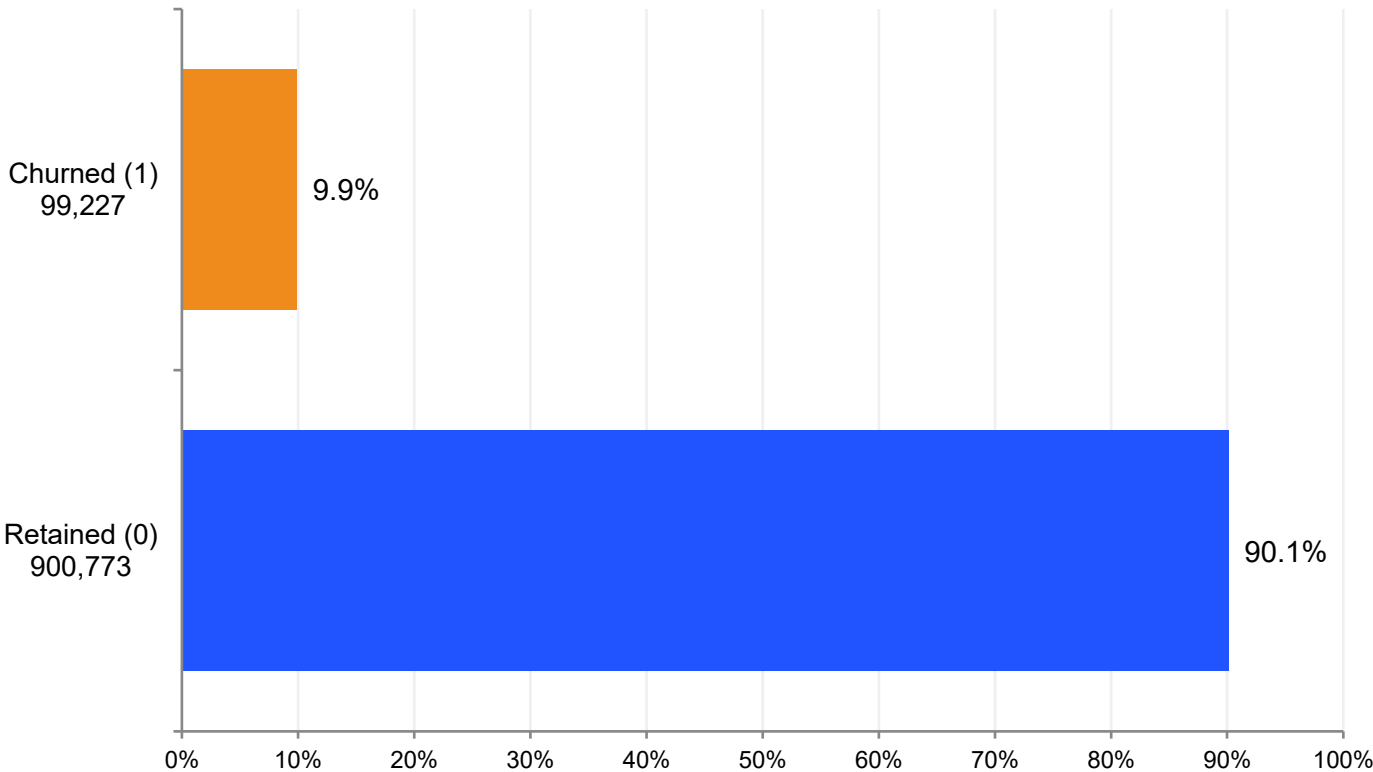
Total Rows: 1,000,000
Features: 32 columns (24 numeric / 8 categorical)
Task Type: Binary Classification
Duplicate Rows: Zero Detected

⚙️ PROPOSED APPROACH

- Exclude customer_id; engineer tenure from signup_date
- Encode categorical contract/payment fields
- Compare RF/XGBoost vs Logistic baseline
- Evaluate via ROC-AUC / recall (9.9% churn rate)

MISSINGNESS RISK
avg_monthly_gb: 5.0%
credit_score: 4.0%

Target Distribution: churn (binary)



Imbalanced target — plan: stratified split + class weights + precision/recall over accuracy

OUTLIER IQR FLAGS
avg_monthly_gb: 56,428
totalcharges: 45,654

LINEAR SIGNAL
Weak (max |r| = 0.085)
customer_satisfaction ↓

Thank you ♥

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